

JAIDEEP SAHU

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Professional Summary

Backend Software Engineer with 2 years of experience developing and testing Python-based backend services and REST APIs. Skilled in Python, FastAPI, Django, PostgreSQL, Pytest, debugging, performance optimization, and cloud deployments on AWS and GCP. Experienced in troubleshooting production issues and improving the reliability, scalability, and performance of backend systems.

Skills

- **Languages:** Python, Java
- **Frameworks:** FastAPI, Django, Django REST Framework, Bottle
- **Testing & Quality:** Pytest, Unit Testing, Debugging, API Testing, Production Troubleshooting
- **Databases:** PostgreSQL, InfluxDB
- **Cloud & Tools:** AWS, GCP (Cloud Run), Docker, Linux, Gunicorn, Uvicorn, Git, GitHub, Bitbucket, Jira, Postman, VS Code
- **Architecture & Practices:** REST APIs, Microservices, Server-Sent Events (SSE), WebSockets, SDLC, STLC
- **Generative AI:** LLM Integration, RAG, Anthropic Claude, Prompt Engineering
- **Soft Skills:** Cross-functional Team Collaboration, Adaptability, Analytical Problem Solving

Experience

Software Engineer — Trinesis Technologies Pvt Ltd. — August 2024 - Present

- Designed and developed backend systems for dashboards supporting operational data visualization
- Designed and optimized PostgreSQL database architecture for efficient data storage and retrieval
- Developed Django REST Framework APIs with routing, views, and serializers for seamless system integration
- Developed and maintained unit tests using Pytest and performed debugging to improve backend service reliability
- Collaborated with cross-functional teams using Django MVT architecture to design scalable systems
- Improved backend performance through connection pooling and SQL query optimization

Projects

DRIMS (Digital Record Information Management System) — Client: Brilliant Corporation

- Developed two Python backend services and enhanced existing services by resolving critical bugs, adding features, and improving overall platform performance
- Replaced the default Bottle development server with Gunicorn for selected services, enabling concurrent request handling and reducing container downtime
- Reduced peak memory usage of the TIFF-to-PDF conversion service by roughly 93% (3.8 GB to 250 MB) and output PDF size by 90% (76 MB to 8 MB) by replacing Pillow with img2pdf, eliminating memory-related request failures
- Diagnosed and fixed a production issue where memory exhaustion blocked database-entry creation, causing a batch daemon to repeatedly re-upload the same files; prevented duplicate AWS Batch jobs and stabilized the pipeline
- Deployed, monitored, and troubleshot cloud-hosted services on AWS, improving system stability and scalability

FLUXJIVA

- Built backend services for FLUXJIVA, an on-machine application for real-time monitoring of live machine data with an integrated AI assistant
- Developed the core chat service in FastAPI with RAG over company knowledge and live machine data, invoking Anthropic Claude Sonnet and deploying the service on GCP Cloud Run
- Migrated real-time communication from WebSockets to Server-Sent Events (SSE), eliminating connection timeouts under heavy data load and enabling efficient streaming to multiple concurrent clients
- Reduced InfluxDB CPU utilization from over 90% to a 20–40% average by implementing a caching layer for OEE calculations, reducing request failures under load
- Built secure user authentication and token management using Django's authentication system

Education

Parul Institute of Technology, Parul University, Vadodara, India

2019 - 2023

- Bachelor of Technology (B.Tech) in Computer Science Engineering